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Miocene remagnetisation of carbonate platform and Antalya Complex units within the Isparta Angle, SW Turkey

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ABSTRACT

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A palaeomagnetic study has been carried out within the Mesozoic and Tertiary units of the relatively autochthonous carbonate platforms and the allochthonous deep-sea volcanics and sediments of the Antalya Complex, exposed around the Isparta angle, SW Turkey. The Antalya Complex is interpreted as a mosaic of carbonate platforms, basinal sediments, volcanic and ophiolitic rocks which formed within a southerly strand of the Neotethyan ocean, adjacent to Gondwana.

The results indicate a widespread remagnetisation event. Negative fold tests show that the remanence at most sites is of secondary origin (e.g., within the Çirali lavas). The magnetisation is carried by magnetite of presumed authigenic origin. The remagnetisation event is believed to have occurred in the Early–Middle Miocene (Burdigalian–Langhian). It was possibly triggered by the migration of orogenic fluids ahead of the advancing Lycian nappes during their emplacement onto the carbonate platforms.

Subsequent to remagnetisation, a large segment of the Isparta angle underwent an anticlockwise rotation of 30°. This rotation is attributed to the overall convergence and bending of the Hellenic arc and the final stages of emplacement of the Lycian Nappes during the Late Miocene, in agreement with previous studies.

Previously, southerly palaeolatitudes were inferred from Late Triassic extrusives of the Gödene Zone (Çalbali Dag unit). The post-folding magnetisation identified here within the Cirali lavas of the Gödene Zone to the south implies that these low palaeolatitudes result from the inappropriate application of structural tilt corrections. The available data cannot be used to substantiate an origin for the Antalya units south of the equator in the early Mesozoic. Instead, a position close to the northern margin of Gondwana is indicated.

Introduction

The last decade has seen a substantial increase in the number of studies attempting to use the palaeomagnetic technique as a tool in unravelling the complex tectonic history of mobile belts (e.g., Westphal et al., 1986; Kissel and Laj, 1988; Saribudak et al., 1989). Such studies must necessarily involve suitable field stability tests to determine the timing of acquisition of components of magnetisation identified by detailed demagnetisation experiments.

One of the most intensively studied regions has been the eastern Mediterranean segment of the Tethyan (Alpine–Himalayan) orogenic belt. In particular, palaeomagnetic data from Turkey have been used to suggest that this area was subdivided into three large microplates during the Mesozoic–Early Tertiary. These inferred units underwent rotations and large-scale northward drift from south of the equator since early Mesozoic time (Lauer, 1981, 1984; Fig. 1). As discussed by Robertson and Dixon (1984), however, these inferences from preliminary palaeomagnetic data are not obviously in accord with the geological

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evolution of the area, and there is thus an urgent need to test Lauer's (1981, 1984) hypothesis with additional palaeomagnetic data.

The Isparta angle area of SW Turkey was chosen for such detailed palaeomagnetic study, as this area marks the suture zone of two of Lauer's supposed microcontinents (Fig. 1). According to recent integrated geological studies, the Isparta angle marks the suture zone of a number of microcontinents and small ocean basins (Waldron, 1984a,b; Robertson and Dixon, 1984). For other workers, however, the Isparta angle area is simply part of a much larger, regionally extensive Tauride continental unit, which was overthrust by a set of far-travelled thin-skinned thrust sheets, the "Antalya Nappes". In this hypothesis these "nappes" were envisaged as having been thrust from a Neotethyan ocean basin either to the south (Dumont et al., 1972), or to the north (Ricou et al., 1979).

The initial objective of this palaeomagnetic study was to determine if significant crustal-scale rotations between individual, relatively autochthonous carbonate platforms had taken place at any time during the Early Mesozoic–Tertiary rift/drift/collision history of the area. In practice, it was found that the Tauride carbonate platforms and overthrust units in the Isparta angle area have experienced a widespread Neogene remagnetisation. This has affected units from which inclination data were used by Lauer (1981, 1984) to infer Early Mesozoic southerly palaeolatitudes.

Geological setting

The allochthonous Antalya Complex ("Antalya Nappes") are a structurally complex association of Palaeozoic (Ordovician–Permian), Mesozoic and Early Tertiary sedimentary, igneous and minor metamorphic rocks (Gutnic et al., 1979; Robertson and Woodcock, 1982; Fig. 2). These units regionally overlie large carbonate platforms, known as the "Tauride autochthons". The structurally lower units of the Antalya Complex mainly comprise Mesozoic basinal sediments, mafic extrusives and ophiolitic rocks. Structurally higher units are mainly Mesozoic platform carbonates. In some areas, however, this simple structural order breaks down, for example where basinal and platform units are structurally interleaved, or where renewed thrusting has disrupted the initial tectonic stacking order.

In general, the regional basement of relatively autochthonous carbonate platform units comprises Mesozoic platform sediments, overlain by Late Cretaceous pelagic carbonates. These sediments are depositionally overlain by Maastrichtian? to Palaeocene–Early Eocene? clastic and pelagic sediments containing ophiolite detritus related to the emplacement of oceanic crust from the Neotethyan ocean. Platforms around the eastern edge of the Isparta angle are exposed from the base upwards (e.g., Anamas Dag; Fig. 2), while only the higher stratigraphic levels (Cretaceous and younger, Tertiary) of the large platform are seen around the southwestern mar-

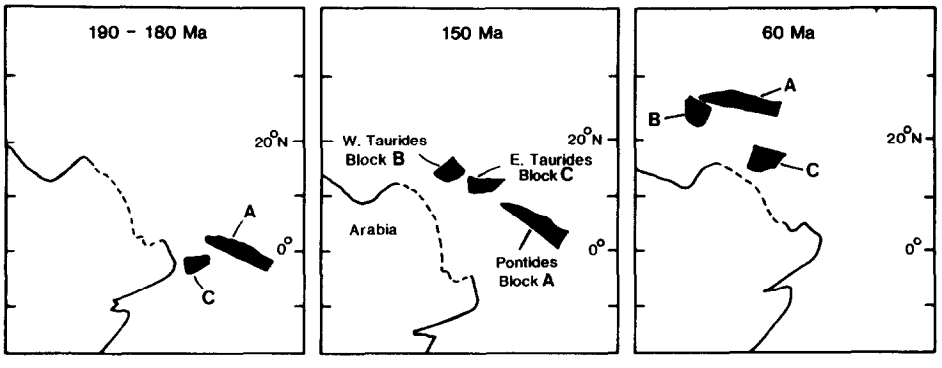


Fig. 1. Model of the separation of Anatolia into three microplates, *A*, *B* and *C*, which drifted northwestwards from an initial position off SE Arabia (modified from Lauer, 1984).

gin of the Isparta angle (Bey Daglari; Fig. 2). Most of the remaining carbonate platforms are demonstrably allochthonous and are mainly located in a high structural position. Particularly notable are the Kemer units of the southwestern area, in which long, intact successions of Ordovician–early Tertiary age are exposed. Also significant is the Barla Dag platform in the north of the Isparta angle, where an unusual condensed, Ammonitico Rosso facies of Early Jurassic–Early Cretaceous age is developed (Gutnic et al., 1979). This platform unit shows no evidence of large-scale overthrusting during the regional emplacement of allochthonous units (Robertson et al., 1991; Robertson, 1992) and was the only unit sampled not to show evidence of remagnetisation. Most of the allochthonous slope and basinal units are too highly deformed to be considered worth palaeomagnetically sampling. However, particularly in the southwestern area, thick relatively intact successions of Late Triassic pillow basalts are well exposed (e.g., Çalbali Dag; Juteau, 1975; Fig. 2). These extrusives were erupted during the early stages of Neotethyan spreading in this area (Juteau, 1975; Robertson and Woodcock, 1981c).

The main structural events and their interpretation are detailed in Table 1.

Previous palaeomagnetic research

Lauer (1981, 1984) has carried out sampling of carbonates and igneous units of the Antalya Complex and underlying relative autochthons as part of a study and review of Turkish palaeomagnetism. This work was aimed at placing constraints of the positions of the various Turkish blocks during the Mesozoic and Tertiary (Fig. 1). Four sites are described by Lauer (1981) from within the uppermost (Maastrichtian to Palaeocene) sequences of the Bey Daglari carbonate platform. Both normal and reversed polarities were found. The generally NW-directed declinations (after normalising reversed polarity samples) indicated a significant anticlockwise rotation of the sampled units, although within-site dispersion was high at two sites ($\alpha_{95} > 20°$).

TABLE 1

The main structural events and their interpretation

	Period/Epoch	Structural event	Interpretation
1.6	Pliocene/ Pleistocene	Extension and/or strike-slip Kovada graben	Reactivation of Antaya Complex suture zone
5	Miocene	Compression in SW, W, E, and SE areas	Tightening of suture due to advanced collision elsewhere
23	Oligocene	Extension	Post-suture Aksu clastic basin
35	Eocene	NE–SW thrusting in NE area only, NW–SE thrusting in N area	Re-thrusting due to collision N
50	Palaeocene	Large-scale overthrusting	Collision and suturing of platforms and basins
65	U Cretaceous L	Extensional faulting, subsidence	Supra subduction zone spreading Neotethys
145	U Jurassic M L	Extensional faulting, volcanism Gentle subsidence	Renewed seafloor spreading Relative stability
208	U Triassic M L	Extensional faulting, volcanism	Initial seafloor spreading Rifting of Neotethys

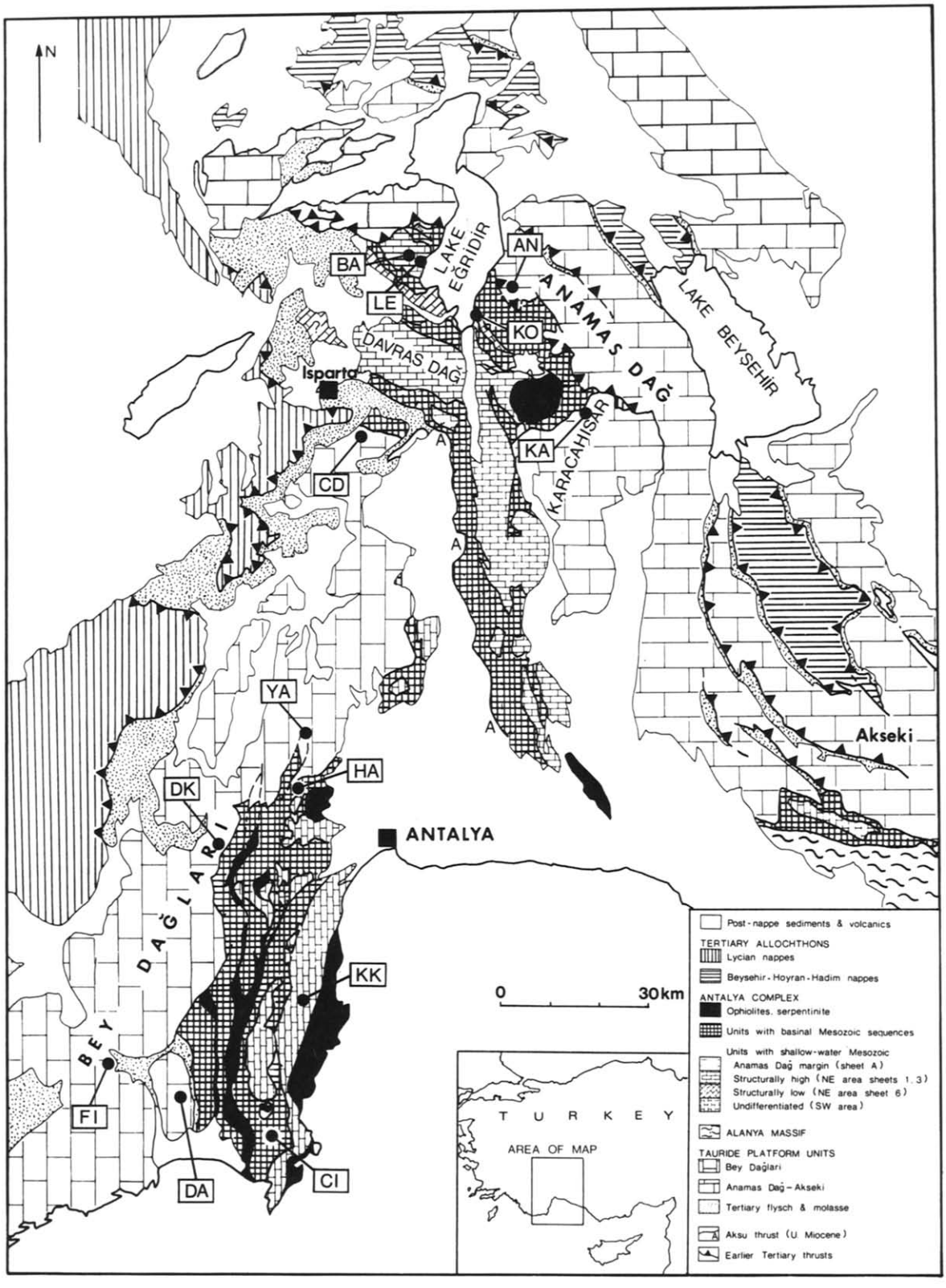


Fig. 2. Geological map of Isparta angle showing the location of the palaeomagnetic sampling sites (modified from Waldron, 1984a).

Eleven further sites were located in Late Triassic pillow lavas of the Çalbali Dag massif (Lauer, 1981), within the Gödene Zone of Robertson and Woodcock (1981c). NW-directed declinations were also found at these sites, but with shallow negative inclinations. These data were used by Lauer (1981, 1984) to infer large-scale northward drift of his Turkish Block *B* from south of the equator since the early Mesozoic. These data are reinterpreted below in the light of the findings of the present study.

Kissel and Poisson (1986) report data from ten sites located within Langhian (Middle Miocene) to Late Pliocene marls and carbonates, Early Pliocene lavas and Pleistocene ignimbrites of the Antalya basin. Both normal and reversed polarities of magnetisation were found, and a positive fold test indicated that the characteristic magnetisation at the sites predates folding. The mean direction of magnetisation indicated that the sampled sequences of the Antalya basin were not subjected to any significant rotation since the Langhian.

Kissel and Poisson (1987) have also studied 17 sites in sedimentary formations on the western flank of the Bey Daglari carbonate platform, and six sites within the Eocene carbonates of the Lycian Nappes. Data from the latter sites were rejected, but reliable results were obtained from seven sites of Eocene–Early Miocene age on the Bey Daglari. Again, both normal and reversed polarities were found, but no fold test was possible. The mean direction of magnetisation, both before and after correction for bedding tilt, indicated an anticlockwise rotation of the Bey Daglari of approximately 30° (with respect to present north). The timing of this rotation is constrained to be post-Early Miocene by the age of the sampled units (Kissel and Poisson, 1987). This result was in agreement with the preliminary data of Lauer (1981).

Sampling and methods

All sites were located at palaeomagnetically unstudied exposures. Samples were collected from the relatively autochthonous Bey Daglari carbonate platform in the southwest, the Anamas Dag and Karacahisar carbonate platforms in the east and the Barla Dag to the north of the Isparta angle (Fig. 2). In addition, the relatively allochthonous, structurally higher carbonate platforms of the Davras Dag and Sutculer-Dulup units were sampled. The sites were located in structurally coherent, well-dated carbonate platform and pelagic carbonate successions (Poisson, 1977; Akbulut, 1977; Gutnic et al., 1979; Robertson and Woodcock, 1982; Waldron, 1984a,b). Although some sites were located in Triassic, Jurassic and Early Cretaceous sections, sampling was concentrated in the Late Cretaceous–Early Tertiary stratigraphic interval for two main reasons:

(1) Late Cretaceous–Early Tertiary pelagic successions are thinly-bedded and fine-grained, in contrast to many earlier Mesozoic platform limestones, which are massively bedded, coarse-grained and recrystallised. The regular bedding planes of the pelagic facies allowed the accurate determination of appropriate structural tilt corrections at each site. Also, fine-grained sediments are generally considered to be more useful for palaeomagnetic studies.

(2) Geological studies suggest that important tectonic rotations are likely to have occurred between independent carbonate platform units during the Late Cretaceous and Early Tertiary phase of crustal convergence and collisional tectonics (e.g., Waldron, 1984a,b), rather than during the Triassic to mid-Cretaceous passive margin phase.

Additional samples were taken from the allochthonous basinal units of the Antalya Complex, especially within the well documented exposures of its southwestern segment, adjacent to the Bey Daglari carbonate platform (Fig. 2). Here, sites were located: (1) in Late Triassic extrusives, interpreted as marginal oceanic crust (Gödene Zone), and (2) within a complete Ordovician to Late Cretaceous section (Kemer Zone), interpreted as a small off-margin platform (Robertson and Woodcock, 1980, 1981a,b,c, 1982, 1984).

Samples were drilled in the field by T.M.M. Clube (Clube, 1986). Each sample was orientated with both magnetic and sun compasses (Creer and Sanver, 1967). Visibly weathered material was avoided, together with sections close to faults, landslides and zones of penetrative veining.

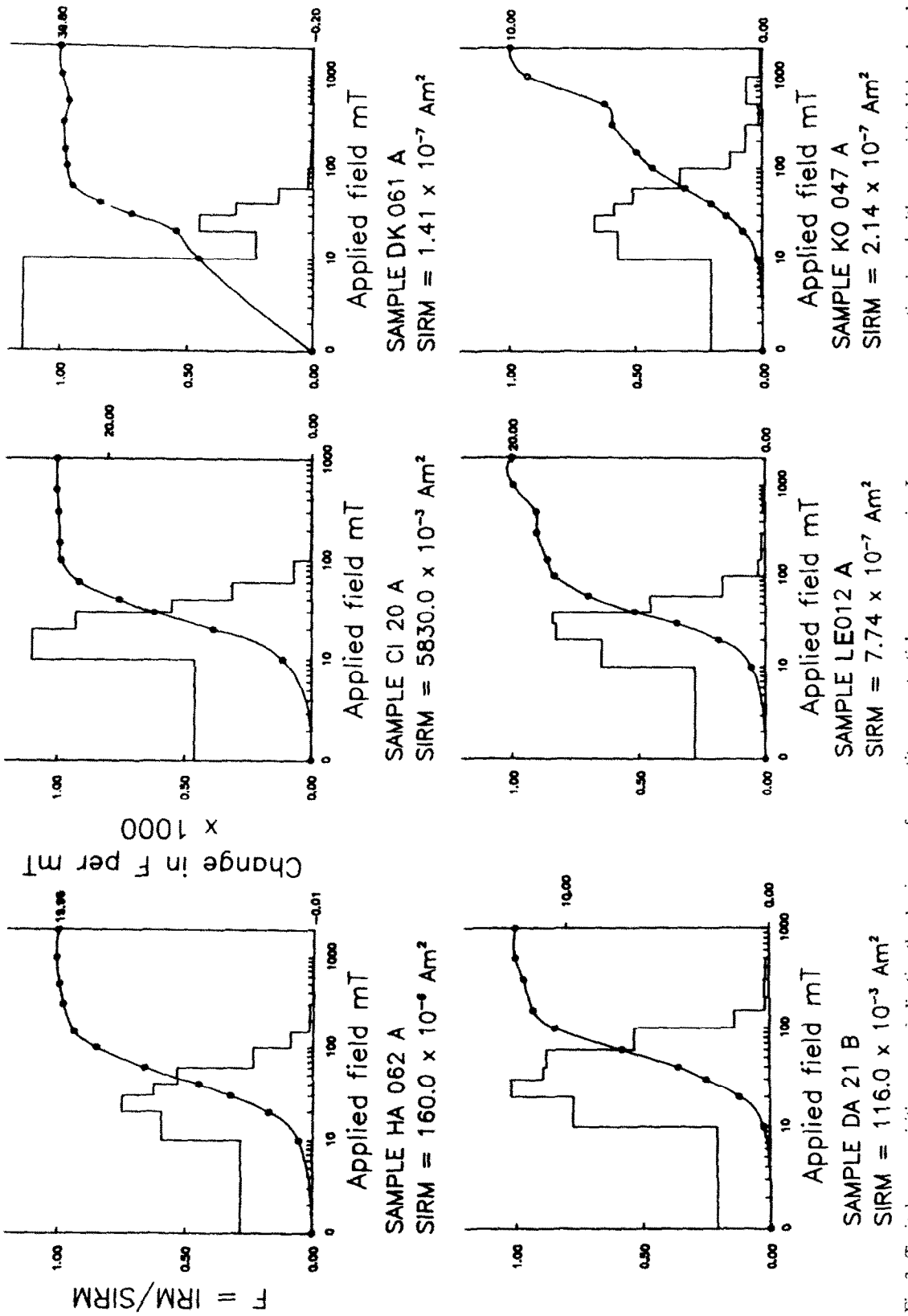


Fig. 3. Typical IRM acquisition curves, indicating the dominance of magnetite as a potential remanence carrier. In some cases a magnetic mineral with a coercivity higher than that of magnetite is also present.

In the laboratory, samples were analysed using a 2-axis CCL superconducting magnetometer. The NRM intensities of many of the carbonate samples were at or close to the limit of sensitivity of the cryogenic magnetometer (approximately 5.0×10^{-6} Am^{-1}). Even when statistically acceptable NRM measurements were obtained, many samples became too weakly magnetised to measure after the initial stages of demagnetisation. Measurements carried out on other samples frequently passed the statistical rejection criteria based on the sample "internal" standard deviation (Lowrie et al., 1980), but showed large variations in remanence direction between successive demagnetisation steps. Thus, a large number of samples were rejected at an early stage in the analysis. Good quality demagnetisation data were obtained from 13 sites, and only those samples which yielded stable endpoints have been included in the site mean directions given below. The location of these sites is shown in Figure 2.

Magnetic properties

The limestones of the Isparta angle are in general very weakly magnetic. The Tauride platform carbonates have a mean NRM intensity of 7.6×10^{-5} Am^{-1}, and the platform and passive margin carbonates of the Antalya Complex have a mean intensity of 5.4×10^{-5} Am^{-1}. Higher intensities were found within the Ordovician to Late Cretaceous section (mean = 2.3×10^{-3} Am^{-1}). No attempt has been made to obtain magnetic extracts from these carbonate samples because of the low concentration of magnetic minerals implied by these intensities. In general, separation techniques which have been tried by other workers on such carbonates have only been partially successful in enriching a pure and representative sample for thermomagnetic analysis (Lowrie and Heller, 1982, and references therein). Instead, the magnetic mineralogy of the sampled lithologies has been determined from the coercivity and blocking temperature characteristics of both natural and laboratory-grown remanences. Additionally, one site was located in the Late Triassic lavas of the Gödene Zone (Robertson and Woodcock, 1981c), at Çirali (site CI; Fig. 2). The mean NRM intensity of samples from this site was 8.7 Am^{-1}. Measurements were made on these samples using a Molspin spinner magnetometer.

The rate of acquisition of isothermal remanence (IRM) in fields up to 1.0–2.0 T has been studied for at least two samples per site. Typical results of the IRM analyses are shown in Figure 3. In the majority of samples, a rapid initial rise and subsequent flattening off of curves below 0.3 T indicates that magnetite is the main magnetic mineral present. In some cases, however, a continuous increase in the IRM for fields higher than 0.3 T is observed. This indicates the presence of a higher coercivity mineral.

To confirm the presence of magnetite and to identify the higher coercivity fraction, one sample from each site was given a composite IRM (2.0 T IRM along the $-Z$ sample axis, and a 0.3 T IRM along the $+Z$ axis) and then stepwise thermally demagnetised, following the example of Channell et al. (1982). Figure 4 shows typical normalised intensity against temperature curves. Figure 4a illustrates typical curves found in those samples where only magnetite was identified by the IRM acquisition analyses. The IRM grown at 0.3 T along the $+Z$ sample axes, which affected only the low-coercivity grains, is removed continuously up to temperatures of 520–580°C. These unblocking temperatures are consistent with the IRM being carried by relatively titanium-free titanomagnetite.

More complicated demagnetisation paths are observed in those samples where a high coercivity mineral was identified by the IRM analyses (Fig. 4b). In these cases, the 0.3 T IRM is preferentially destroyed up to temperatures of about 550°C. At approximately 400°C, the normalised intensity reaches zero, indicating that the remaining intensities of the two antiparallel IRM's cancel out. The 0.3 T IRM is completely destroyed at 550°C, when minima in the curves are observed. Again, this indicates that these IRM's are carried by relatively pure titanomagnetite. At temperatures above 550°C, only the higher-coercivity (2.0 T) IRM's remain. These are destroyed by approximately 650°C, indicating a haematite carrier.

The domain state of the titanomagnetite carriers, identified by both the IRM acquisition and

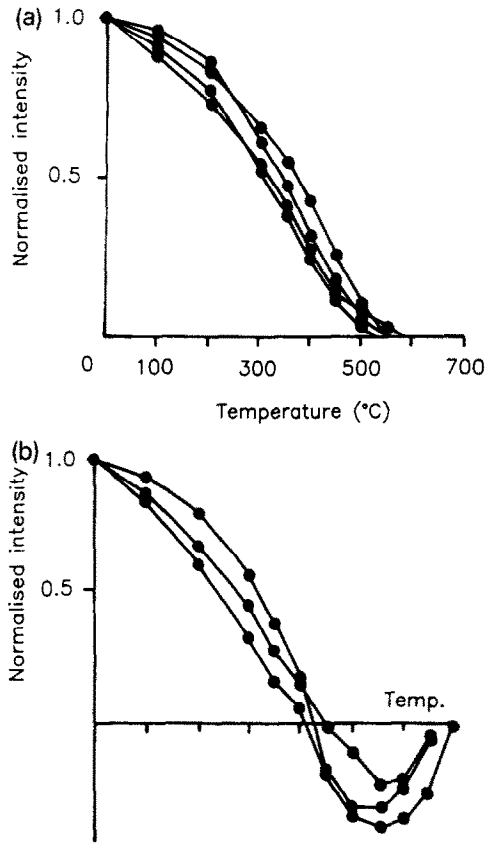


Fig. 4. Thermal demagnetisation of composite IRM's, following the example of Channell et al. (1982). Low coercivity carriers have a positive magnetisation, and high coercivity carriers are negative. (a) In the majority of samples, a continuous reduction in intensity up to maximum unblocking temperatures of 520–580°C confirms the presence of magnetite as the dominant magnetic mineral present. (b) In those cases where a higher coercivity mineral was identified by the IRM analyses, a decrease in normalised intensity up to temperatures of 550°C indicates the presence of magnetite, whereas maximum unblocking temperatures of approximately 650°C for the 2.0 T IRM indicate that haematite is also present.

thermal demagnetisation experiments, has been examined by determining the relative resistance of laboratory-induced isothermal (IRM) and anhysteretic (ARM) remanences to alternating field (AF) demagnetisation (Johnson et al., 1975). In every instance the weak-field ARM was found to be more resistive to AF demagnetisation than the IRM, indicating that single-domain-sized grains dominate the magnetite fraction of the samples.

Stepwise AF demagnetisation of a minimum of ten samples per site was carried out up to peak fields of 100 mT. In many cases, the intensity of magnetisation decreased below a level at which accurate measurement could be made (approximately 10.0×10^{-6} Am^{-1}) before final endpoints were obtained. In samples where stable components of magnetisation were recovered, AF demagnetisation was found to be effective in removing all the magnetisation. This indicates that the haematite fraction identified by the thermal demagnetisation of composite IRM's does not carry any significant natural remanence. Typical demagnetisation diagrams are shown in Figure 5. A common N-dipping component, attributed to viscous magnetisation in the present field direction, was removed by fields of less than 15–20 mT to leave single stable components of magnetisation.

A limited number of samples was subjected to thermal treatment to investigate the blocking temperature spectra of the natural remanence carried by the platform carbonates. Two examples of the Zijderveld demagnetisation diagrams obtained are shown in Figure 5. Maximum unblocking temperatures of between 350 and 500°C were found (Fig. 6). These data, in conjunction with the effectiveness of AF demagnetisation in reducing NRM intensities to zero, demonstrate that only magnetite carries a natural remanence in these samples.

In summary, the rock magnetic experiments indicate that fine-grained magnetite carries the characteristic magnetisation in all the samples studied. Maximum unblocking temperatures of between 350 and 500°C were found. Haematite, where present, does not contribute significantly to the natural remanence.

Palaeomagnetic results

Relatively autochthonous carbonate platform units

Reliable palaeomagnetic results with directions unrelated to the present geomagnetic field direction were obtained from nine sites. Only data from those samples which showed stable demagnetisation paths were used in calculating site mean directions. The directions of the cleaned remanence vectors were found by drawing best-fit lines through the last four or more points on

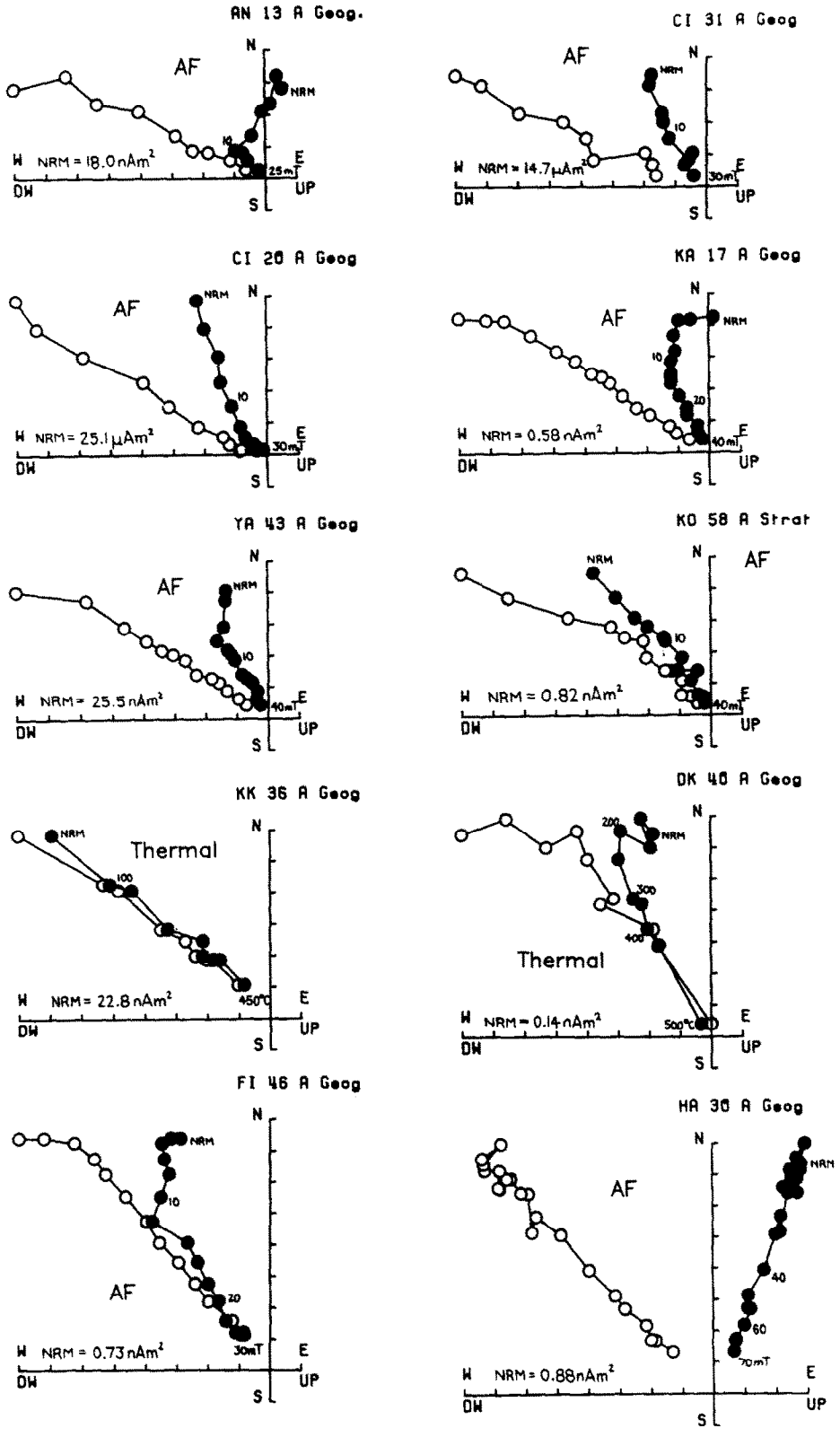


Fig. 5. Examples of both alternating field (AF) and thermal demagnetisation. Note the dominance of NW-directed declinations, the positive inclinations of all samples, and that AF treatment is effective in removing all the natural remanence. All samples are shown prior to tilt correction, with the exception of sample KO 58A. (● = horizontal plane, ○ = vertical N–S plane. Note that the vertical projections are on the horizontal axes).

demagnetisation diagrams. The cleaned site mean directions are reported, both before and after the application of standard structural tilt corrections (Table 2).

At five sites within the Bey Daglari carbonate platform (Table 2), the in situ characteristic remanence directions cluster in the NW quadrant. At some of these sites, demagnetisation has resulted in a slight increase in within-site dispersion compared with the NRM directions. An example is site YA (Bayatbademlisi), located in the Palaeocene sequence at the top of the Bey Daglari platform succession. The lower dispersion before magnetic cleaning is attributed to the presence of viscous components acquired in the present geomagnetic field at the sites. These viscous compo-

TABLE 2

Palaeomagnetic results

Site	Age	*N*	In situ				Tilt corrected			
			D	*I*	α_{95}	K	*D*	*I*	α_{95}	K
RELATIVELY AUTOCHTONOUS CARBONATE PLATFORM UNITS										
Bey Daglari										
DA	Palaeocene	4	314	27	13.2	50	344	44	14.7	40
FI	Palaeocene	12	316	50	7.2	38	354	40	7.9	30
DK	Palaeocene	8	334	20	12.3	21	349	45	12.3	21
YA	U Cre–L Tert	27	313	50	7.0	17	312	39	7.4	15
CD	U Cre–L Tert	5	325	50	10.2	57	352	42	16.9	21
Barla Dag										
BA	U Triassic	9	323	19	9.4	31	320	25	12.1	19
LE	Pal/Eoc flysch	6	331	49	10.0	46	335	29	10.0	46
Anamas Dag										
AN	Pre–mid Jur	13	331	56	4.1	103	320	68	6.9	37
Karacahisar platform										
KA	U Cretaceous	4	340	47	13.6	47	015	17	13.6	47
ALLOCHTHONOUS ANTALYA COMPLEX UNITS										
Passive margin sequence of Bey Daglari										
HA		37	033	47	4.5	29	037	12	6.9	13
Çayköy Ridge										
KO	U Cre	7	349	64	8.9	47	328	44	8.9	47
Gödene Zone										
CI	U Triassic	18	334	49	5.6	40	309	−46	11.8	10
	Sub-site 1	7	328	45	10.7	32	325	−20	10.7	32
	Sub-site 2 (overturned)	11	338	51	6.4	52	289	−60	6.4	52
Kemer Zone										
KK	Ord–U Cre	123	321	36	3.2	17	056	34	5.4	7
	Ordovician	27	320	39	7.9	13	112	29	7.9	13
	Permian	49	321	36	4.6	21	049	26	5.3	15
	Triassic	7	305	56	22.0	8	055	20	22.0	8
	Jur–U Cre	40	322	29	4.6	25	030	40	4.5	27

N = number of samples; α_{95} = semi-angle of 95% cone of confidence; K = Fisher precision parameter

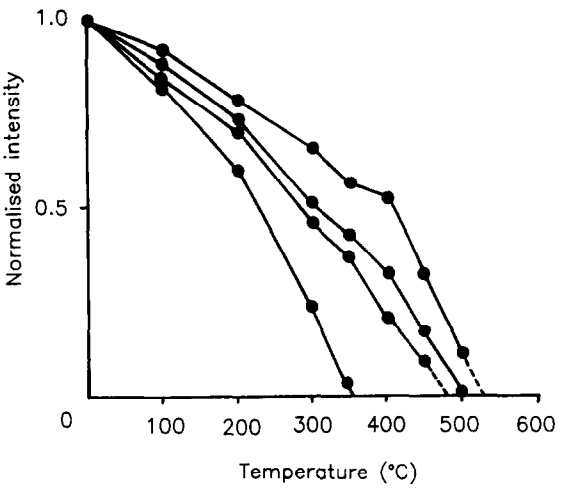


Fig. 6. Thermal demagnetisation decay curves for various platform samples. The maximum unblocking temperatures are between 350 and 500°C. Together with the AF demagnetisation data, these data indicate that only magnetite carries a natural remanence in these samples.

nents have a biasing effect on the sampling of secular variation of the geomagnetic field inherent at each site.

At site CD, located in the upper part of the Bey Daglari platform succession at Çamlidere, tilt correction results in an increased dispersion and a movement of the site mean direction away from a NW azimuth (Table 2). The increase in dispersion could indicate that the magnetisation of these samples was acquired after folding at the site, but this inference is not statistically significant.

More conclusive evidence for a post-folding age for the magnetisation identified within the Bey Daglari carbonate platform comes from a comparison of the cleaned remanence directions at sites YA (Bayatbademlisi) and FI (Finike; Fig. 2). At these sites, the in situ sample directions cluster in the NW quadrant with overlapping α_{95} cones of confidence. However, after correction for bedding tilt, directions from the two sites diverge (Fig. 7). This negative fold test is significant at the 95% confidence level (McFadden and Jones, 1981), and indicates that the magnetisation carried by the uppermost sequences of the Bey Daglari was acquired after folding occurred.

The only stable endpoints with reversed polarity identified in this study were found within Late Triassic calciturbidites of the Barla Dag unit (Site BA; Fig. 2). After removal of an important northward dipping viscous component during the initial stage of magnetic cleaning (using both AF and thermal demagnetisation methods on different samples), sample directions at this site formed a normal–reverse polarity couple (Fig. 8). Such antipodal normal and reverse polarity groups are normally cited as indicating the stability and primary nature of the measured remanence at a site (e.g., Kissel and Laj, 1988). However, in view of the secondary nature of the remanences identified at other sites in this study, there are two possible explanations for the observed distribution: (1) the sampled strata were deposited during at least one reversal of the geomagnetic field and the primary remanence has survived; or (2) the magnetisation represents a secondary overprint similar to that identified at other sites, in which case the remagnetisation event must span the time period of at least one reversal.

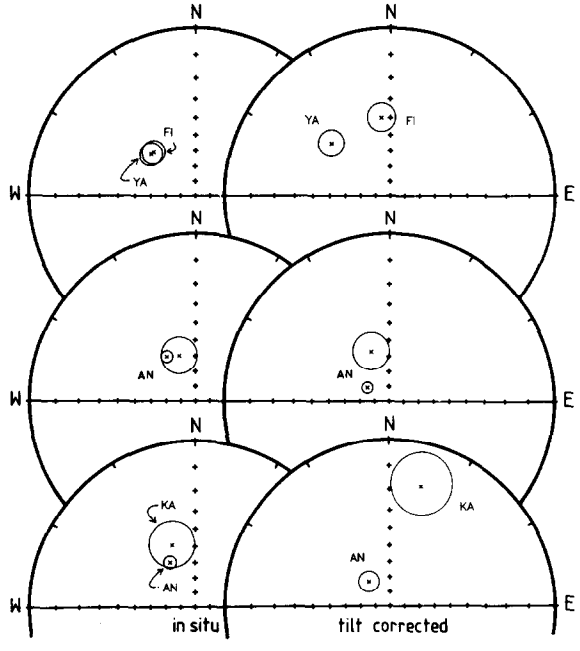


Fig. 7. Negative fold tests at sites within the relatively autochthonous carbonate platform units. Upper stereonet: between-site test at sites FI and YA (uppermost levels of the Bey Daglari platform). Middle stereonet: within-site test at site AN (Anamas–Akseki–Karacahisar platform). Lower stereonet: between-site test at sites AN and KA (Anamas–Akseki–Karacahisar platform). All three tests are significant at the 95% confidence level.

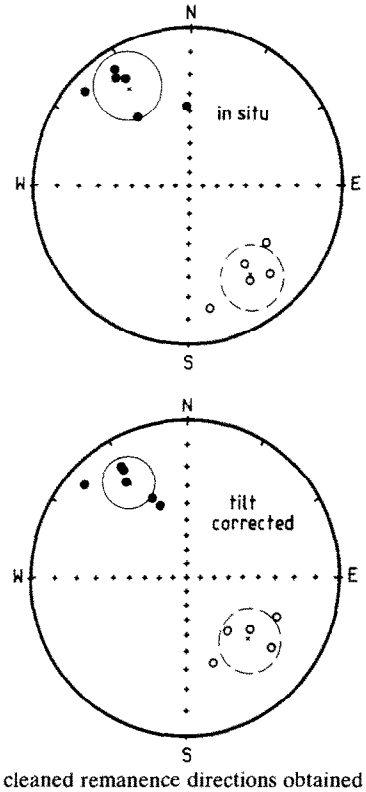


Fig. 8. The cleaned remanence directions obtained at site BA, located within Late Triassic sequences of the Barla Dag carbonate platform. Both before (upper stereonet) and after (lower stereonet) tilt correction, the mean directions of the normal and reversed polarity groups are antiparallel. The normal and reversed polarity groups conform to stratigraphic layering, indicating that the magnetisation is probably of primary origin.

The application of a reversal test at site BA is inconclusive. The mean directions of the normal and reversed polarity groups (after normalising the reversed polarities) are indistinguishable statistically (Watson, 1956), both before and after tilt correction. An increase in the angle between the mean directions of the two groups from 3.6 to 16.8° after removal of the bedding tilt may indicate that the magnetisation is of post-folding age. However, the observation that the normal and reversed polarity groups conform to stratigraphic layering suggests that the magnetisation is primary. It is difficult to see how a remagnetisation event could produce such magnetostratigraphic layering. Additionally, results from site LE (Fig. 2) support a pre-folding age for the magnetisation at site BA. The former site is located in flysch of Palaeocene/Early Eocene age, which rests in situ upon the platform carbonates of the Barla Dag massif (Gutnic, 1977). The in situ α_{95} cones of confidence for the two sites are distinct; they are significantly different at a greater than 95% confidence level (McFadden and Jones, 1981). After tilt correction, however, the two sample populations share a common mean direction. This suggests that primary depositional remanent magnetisations are preserved at both sites. Any valid remagnetisation mechanism proposed to explain the post-folding magnetisations seen elsewhere must be capable of accounting for the preservation of these primary remanences. This point will be discussed below.

Only two reliable sites occur within the carbonate platform sequences on the eastern side of the Isparta angle (Table 2). The first is located in the Late Cretaceous section of the relatively autochthonous Karacahisar carbonate platform (site KA). A limited number of samples at this site yielded high quality demagnetisation data. In situ cleaned remanence directions plot in the NW quadrant, but after tilt correction move to the NE quadrant and become much shallower.

The second site is located in Late Triassic–Early Jurassic limestones of the Zindan unit in the NE area of the Antalya Complex (site AN). This unit was interpreted as the western slope facies of the relatively autochthonous Anamas Dag carbonate platform to the east (Waldron, 1984a,b). The Zindan unit is separated from the Anamas Dag carbonate platform unit by a high-angle fault zone. Again, the in situ cleaned vectors from this site (Table 2) cluster in the NW quadrant. Correcting for bedding tilt makes the site mean inclination steeper than the present field inclination at the site (68°, compared with 56° for an axial geocentric dipole field). Also, within-site dispersion is increased by the tilt corrections, with the cleaned vectors dividing into two distinct groups. This provides a within-site negative fold test (Fig. 7) which is significant at the 95% confidence level (McFadden and Jones, 1981) and indicates a post-folding magnetisation.

The data from sites KA and AN also provide an intra-site negative fold test (Fig. 7), which is

significant at better than the 95% confidence level (McFadden and Jones, 1981).

These results suggest that the post-folding remagnetisation identified in the sequences of the Bey Daglari platform to the west was the result of a widespread event. The negative fold tests from both limbs of the Isparta angle indicate that an area-wide test may be possible. We shall return to this point after examining the palaeomagnetic results obtained from the Antalya Complex.

In summary, the palaeomagnetic results from the relatively autochthonous carbonate platform units of the Isparta angle indicate that a strong secondary overprint has completely replaced the primary depositional remanence at all but one site. This pervasive remagnetisation is post-folding in age. Tectonic implications are discussed further below.

Allochthonous Antalya Complex units

Reliable palaeomagnetic results with directions unrelated to the present geomagnetic field direction were obtained from four sites within the allochthonous Antalya Complex units. The cleaned site mean directions are reported in Table 2, both before and after application of standard structural tilt corrections. Of the four reliable sites, three are located in the southwestern segment of the Antalya Complex.

Site HA exhibits the only NE-directed in situ cleaned remanence directions identified within the Isparta angle area (Table 2). This site is located within Late Triassic, *Halobia*-bearing limestones, which represent the slope facies of the Bey Daglari carbonate platform (Çatal Tepe unit; Robertson and Woodcock, 1982; Fig. 2). The unit is separated from the relatively autochthonous Bey Daglari carbonate platform by a low-angle thrust of Late Cretaceous to Palaeocene age. After structural correction, the cleaned remanence directions form two groups: one with shallow inclinations and NE-directed declinations; and a second with steeper inclinations and E-directed declinations. This may indicate that the observed magnetisation was acquired after folding at the site, in agreement with the post-folding age of the magnetisations identified at the Bey Daglari carbonate platform sites. However, the NE in situ site mean declination is difficult to explain, and suggests that the sampled unit could represent a detached block which has experienced a differential rotation after remagnetisation. More conclusive evidence for the secondary nature of the remanence carried by the Antalya Complex units comes from the remaining two sites in the southwestern area.

Site CI is located within the Late Triassic pillow lavas of the Gödene Zone (Robertson and Woodcock, 1981c) at Çirali (Fig. 2). This site is approximately 50 km to the south of the Gödene Zone sites at Çalbali Dag studied by Lauer (1981, 1984). Samples were collected here from two separate fault-bounded blocks. At one of these sub-sites, pillow shapes indicate that the block has been overturned. The cleaned remanence directions obtained from both blocks cluster in the NW quadrant before correction to the palaeohorizontal (Fig. 9). The mean direction is in close agreement with the in situ directions obtained from the platform units. After correcting for the observed tilt of the palaeohorizontal at each block, the remanence directions diverge into two groups with negative inclinations (Fig. 9). This negative fold test is significant at the 95% confidence level (McFadden and Jones, 1981), and demonstrates the post-folding age of the magnetisation at the site.

The samples at site KK were collected over a 100 m long continuous section within the Kemer Zone, inferred to be an off-margin, carbonate-capped microcontinental sliver (Robertson and Woodcock, 1982). The sampled section covers the complete Ordovician to Late Cretaceous interval. Table 2 gives a breakdown of the number of samples analysed in each time period. A large number of stable endpoints were obtained at this site, partly because of the significantly higher NRM intensities involved. Two lines of evidence point to the secondary nature of the remanence at site KK:

(1) No reversed polarity samples have been identified, even though the sampled section covers such a long time interval. In particular, over one third of the samples analysed are of Permian age, and the Permo-Carboniferous represents a

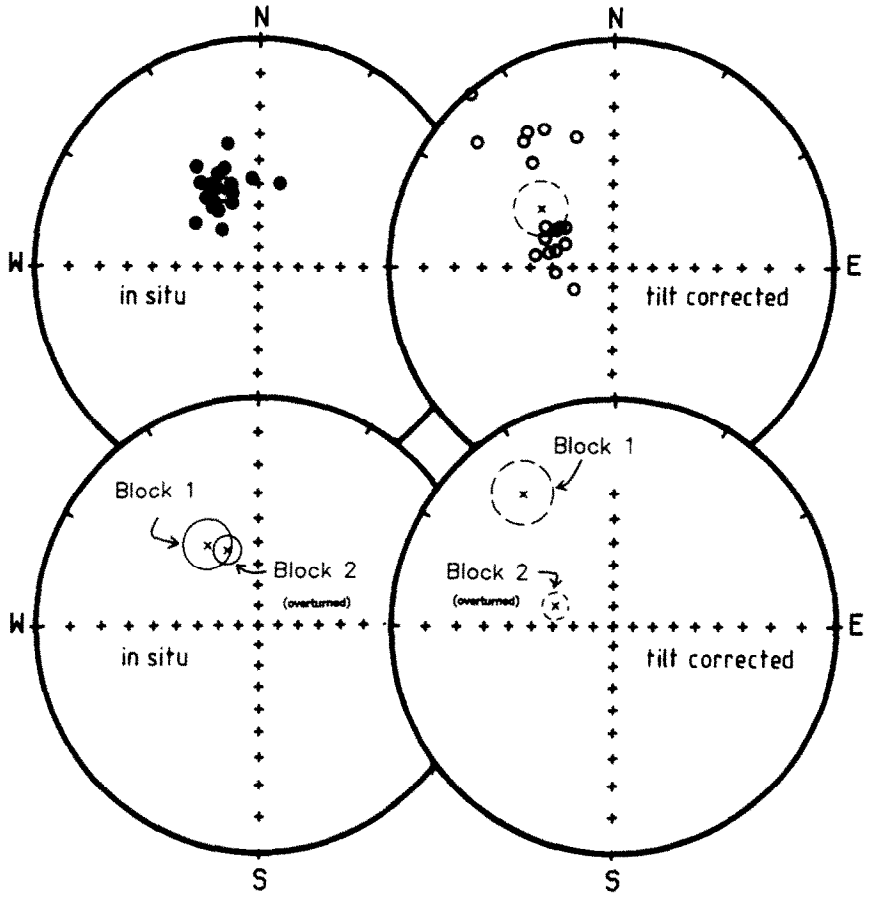


Fig. 9. Upper stereonets: The cleaned remanence directions observed at site CI, within the Late Triassic lavas of the Gödene Zone at Çirali. Lower stereonets: The α_{95} cones of confidence after dividing the data into two groups corresponding to the sampled fault-blocks. Note the divergence of the two mean directions after tilt correction. This negative fold test is significant at the 95% confidence level.

reversed polarity bias superchron (Harland et al., 1982).

(2) The in situ site mean direction is in close agreement with those obtained at site CI and the relatively autochthonous carbonate platform sites. The removal of the stratigraphic tilt along the section results in an increase in the dispersion of remanence directions (Table 2; Fig. 10). If we examine the data in four "time-slices" corresponding to the Ordovician, Permian, Triassic and Jurassic to Late Cretaceous (Table 2), we see that the in situ α_{95} cones of confidence for these groups overlap in the NW quadrant, but become widely separated after tilt correction (Fig. 10). This negative fold test is significant at the 95% confidence level (McFadden and Jones, 1981).

The only reliable site mean direction obtained in the northern outcrop areas of the Antalya Complex allochthonous units comes from site KO (Table 2). This site is situated in the Late Cretaceous section of a small, elongate outcrop of platform carbonates known as the Çayköy Ridge (Fig. 2). This unit has been interpreted by Waldron (1984a,b) as either a continuation of the larger Davras Dag carbonate platform to the west, or as a separate small carbonate platform. The cleaned remanences at this site have an in situ mean inclination which is steeper than the inclination of the present geomagnetic field at the site (64°, compared with 56° for an axial geocentric dipole field). After tilt correction, however, the site mean direction becomes indistinguishable

from the remagnetised directions described at other sites (Table 2). There is a possibility that the magnetisation at this site is a primary depositional remanence. However, this result is interpreted as a secondary remanence acquired prior to the minor tilting at the site (mean dip = 21°). This tilting may be related to Neotectonic faulting of the NE Antalya Complex area.

In summary, these results indicate that the remagnetisation which has affected the relatively autochthonous carbonate platform sequences has also resulted in the destruction of the primary remanence of the sampled units of the relatively allochthonous Antalya Complex. This secondary magnetisation has been shown to reside purely in magnetite.

Summary of evidence for extensive remagnetisation

The evidence for the secondary nature of the measured remanent magnetisations includes:

(1) within-site negative fold tests at site AN in the relatively autochthonous Anamas Dag carbonate platform, and at sites CI and KK in the Antalya Complex (Çirali pillow lavas of the Gödene Zone and carbonate platform of the Kemer Zone, respectively), where either the within-site dispersion is increased or the cleaned remanences divide into two or more well-defined groups upon application of structural corrections,

(2) negative fold tests between sites YA and FI within the uppermost sequences of the Bey Daglari carbonate platform, and between sites

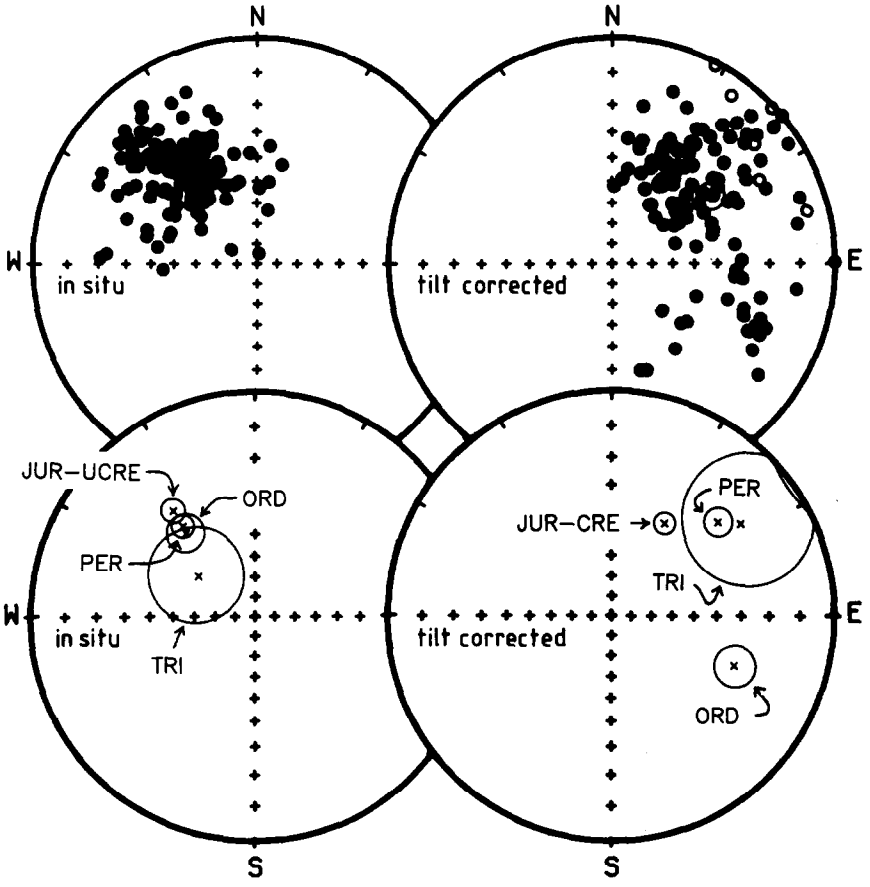


Fig. 10. Upper stereonets: The cleaned remanence directions observed at site KK, which consists of a continuous Ordovician to Late Cretaceous section within the Kemer Zone off-margin carbonate platform. Lower stereonets: The α_{95} cones of confidence after dividing the data into various time-slices. Note the increased dispersion and the divergence of the mean directions after tilt correction. This negative fold test is significant at the 95% confidence level.

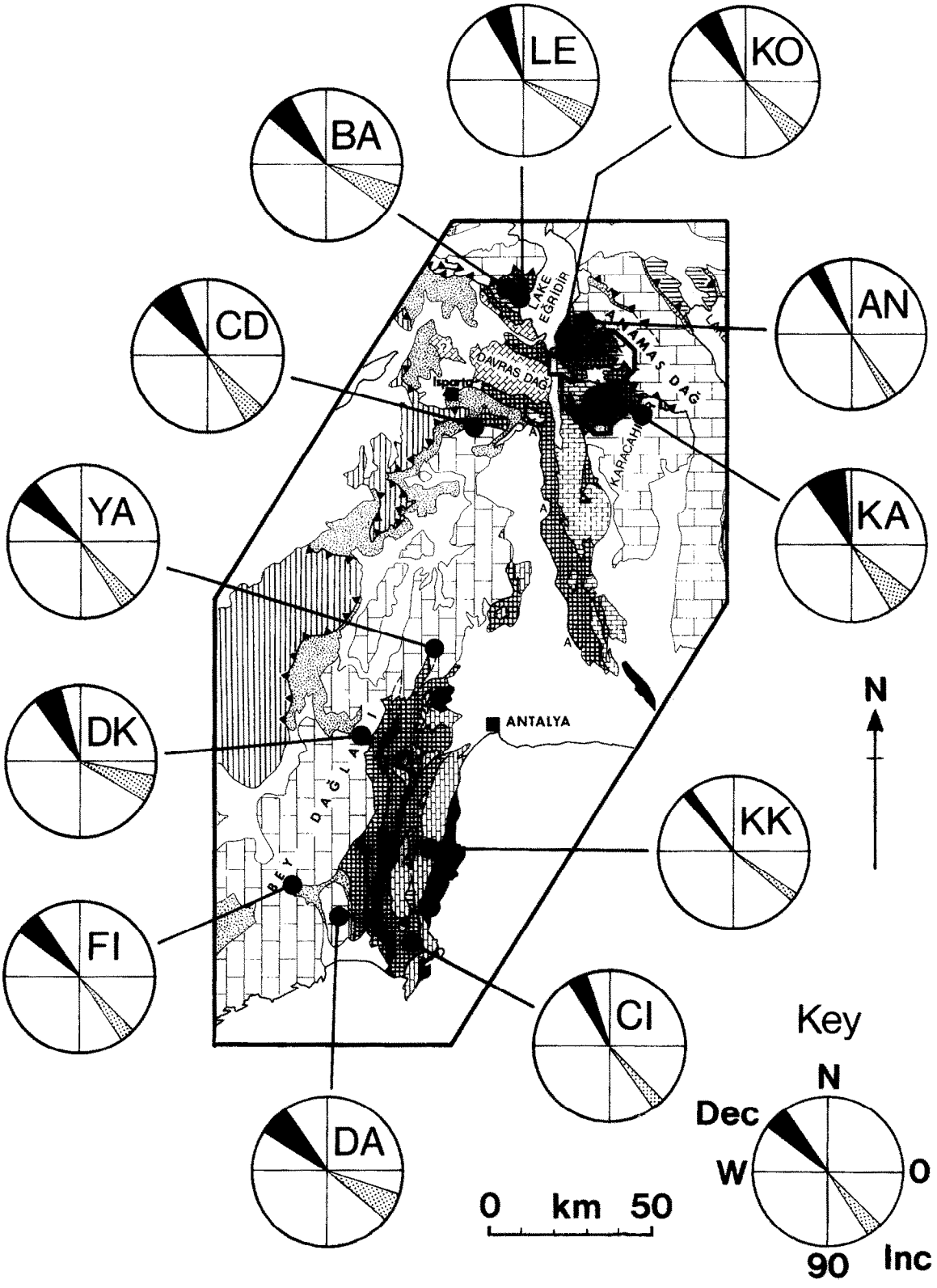


Fig. 11. "Clock" diagrams showing the 95% confidence limits associated with the mean declination and inclination at each site. All data are in in situ coordinates, except for sites BA and LE from the Barla Dag unit (which probably retain primary remanences) and site KO (where post-remagnetisation tilting is believed to have occurred). Data from these three sites are shown after tilt correction. Note the area-wide consistency of the site mean declinations, and the positive inclinations at all sites.

KA and AN in the Anamas–Akseki–Karacahisar platform,

(3) the lack of reversed polarity samples throughout the area, and in particular within the complete Ordovician to Late Cretaceous section at site KK in the Kemer Zone carbonate platform, and

(4) the area-wide consistency of the in situ site mean directions (Fig. 11; Table 2).

The directions of the secondary magnetisation at all sites (Table 2) are unrelated to the present geomagnetic field direction in the study area. The common NW-directed remanence direction must represent an ancient magnetisation acquired during some remagnetisation event. The possible causes of this remagnetisation are discussed below. It should be noted that the sites of Maastrichtian to Miocene age sampled by Lauer (1981) and Kissel and Poisson (1987) in the Bey Daglari carbonate platform may retain primary depositional remanences. These sites record both normal and reversed polarities of magnetisation.

Discussion

Geotectonic interpretation

In the majority of cases, the remanences described above represent secondary magnetisations acquired after tectonic folding, and are characterised by a regional in situ mean direction of: $D = 326°$, $I = 43°$, $K = 37$, $\alpha_{95} = 8.1°$, $N = 10$ sites (note that this mean includes the tilt corrected direction from site KO).

This indicates that, subsequent to remagnetisation, the Bey Daglari platform and the overthrust southwestern segment of the Antalya Complex, the Çayköy Ridge, and at least the northern part of the Anamas–Akseki–Karacahisar platform were all subjected to a ca. 30° anticlockwise rotation (with respect to present north). The youngest units known to have experienced this rotation are Langhian (Early–Middle Miocene) sequences of the Bey Daglari carbonate platform (Kissel and Poisson, 1987). The rotation is therefore of post-Langhian age, and was probably driven by emplacement of the Lycian nappe system in the Middle–Late Miocene, regional convergence and the bending of the Aegean arc (Kissel and Poisson, 1987; Le Pichon and Angelier, 1979; see below).

The close agreement between the declination data at the two Barla Dag sites (BA and LE) demonstrates that no significant tectonic rotation of this massif occurred between the Late Triassic and the Palaeocene/Eocene. The inferred 30–40° anticlockwise tectonic rotation of the Barla Dag platform (with respect to the present north) agrees well with that suggested by the regional mean direction calculated from the remagnetised sites. This implies that the Miocene regional rotation affected both areas to the same extent.

Mechanisms of remagnetisation

A widespread remagnetisation event has occurred within the Isparta angle. The stable, ancient secondary magnetisation is carried exclusively by magnetite, and has completely obscured any pre-existing primary remanence in the units sampled in the Antalya Complex. Components of magnetisation of possible primary origin may be retained by some of the successions of the adjacent relatively autochthonous carbonate platforms.

There are two possible explanations for this pervasive overprint. It represents either: (1) a thermoviscous remagnetisation due to burial and subsequent uplift; or (2) a chemical magnetisation acquired by the growth of authigenic magnetite.

Two main problems are associated with a thermoviscous mechanism. Firstly, thermal demagnetisation experiments carried out on samples from several of the carbonate platform sites reveal maximum unblocking temperatures of NRM of between 350 and 500°C (Fig. 6). The units sampled have only been subjected to subgreenschist facies metamorphism and are unlikely to have been buried to depths greater than 1–3 km. Assuming an average geothermal gradient of 25°C/km, these burial depths suggest that the units have been heated to a maximum of 75°C. The theoretical curves for magnetite of Pullaiah et al. (1975) and Middleton and Schmidt (1982)

indicate that such temperatures could not produce the observed unblocking temperatures. Secondly, it is difficult to see how a burial and uplift event could result in the pattern of remagnetisation observed here, where all sites in the southwestern segment of the Antalya Complex have been completely remagnetised, possibly only parts of the Bey Daglari carbonate platform have been affected, and where the Barla Dag carbonate platform probably retains a primary remanence.

The alternative explanation is that the secondary magnetisation represents a chemical remanent magnetisation (CRM), produced by the precipitation of authigenic magnetite from fluids migrating through the orogen during crustal convergence. It is well known that when continental margins in zones of crustal convergence are buried beneath thrust sheets, fluids containing abundant hydrated minerals are expelled towards the continental interior. Huge volumes of fluid are involved, perhaps one-third to one-half of the volume of the overthrust units (Oliver, 1986). Such fluids may be expected to react with the host lithologies, possibly releasing iron into solution, which may then contribute to the formation of authigenic magnetite. Elmore et al. (1987) have demonstrated that the precipitation of authigenic magnetite may be linked to the migration of hydrocarbon-related fluids. It should be noted in this respect that drilling by the Turkish geological survey (MTA) has established the existence of heavy oil in the core of the Finike Anticline (Bey Daglari carbonate platform). Also, methane (of uncertain origin) is currently escaping from the Tekirova ophiolite (Juteau, 1975). If the orogenic fluids were also hot, then remagnetisation could be due to both chemical and thermal magnetisation processes.

This model would elegantly explain the pattern of remagnetisation observed in this study. Tectonic fluids forced thorough the orogen would be expected to have the greatest geochemical effect upon the overthrust and tectonised Antalya Complex units and the adjacent parts of the platform. Pervasive remagnetisation would be expected to occur within these units. On the other hand, less tectonised and therefore more impermeable parts of the carbonate platforms would experience little or no remagnetisation, and may be expected to retain a primary magnetisation. Sequences of the Barla Dag carbonate platform escaped remagnetisation because this unit was never overthrust by large-scale ophiolitic units (the platform here is covered by Eocene flysch which contains no ophiolite-derived detritus; Gutnic, 1977; Robertson, 1992).

In recent years it has become established that a similar process was responsible for remagnetisation of Palaeozoic carbonates in the Appalachian Basin of eastern North America. In a study of the Devonian Helderberg and Onondaga carbonates of New York State, Jackson et al. (1988) and McCabe et al. (1989) demonstrated that magnetite concentration along the outcrop belt of the formations varied in a regular fashion. A striking correlation was found between the magnetite concentration and the degree of diagenetic illitisation of clay minerals in a Devonian bentonite horizon, which took place during the Alleghenian orogeny (Carboniferous). Jackson et al. (1988) and McCabe et al. (1989) concluded that most of the magnetite in these carbonates is of authigenic origin, and that clay mineral alteration and magnetite authigenesis were coeval, Late Palaeozoic events that were controlled by the same diagenetic factors. They noted that authigenic potassium-feldspars and potassium-bentonites of Alleghenian age had previously been reported in the basin, giving evidence for potassium-rich orogenic fluids. McCabe et al. concluded that, since the illitisation of detrital smectite requires potassium and that the process may release iron, the introduction of exotic, potassium-rich brines may have triggered both magnetite authigenesis and the illitisation of detrital clays within the limestones.

Age of remagnetisation

The secondary remanence carried by the majority of the sampled units in the Isparta angle area is interpreted as a CRM residing in authigenic magnetite, produced by the interaction of tectonic fluids with the host lithologies. The youngest remagnetised units sampled are parts of

the Palaeocene succession of the Bey Daglari carbonate platform (Table 2). The remagnetisation event therefore must be post-Palaeocene in age and is also constrained to postdate folding by the various negative fold tests.

There have been several stages of deformation in the area. The final stages of regional thrusting and emplacement of the Antalya Complex oceanic units onto the bordering carbonate platforms took place in Late Palaeocene–Early Eocene time (Poisson, 1977; Robertson and Woodcock, 1982). The Isparta angle was deformed again, in the Late Eocene in the northeastern area, including the Anamas Dag carbonate platform margin unit (Zindan unit), and more generally in the Miocene (Gutnic et al., 1979; Robertson and Woodcock, 1982; Table 1). This latter event was related to the regional eastward emplacement of the large-scale Lycian nappe system (Poisson, 1977). The Lycian nappes originated in a Neotethyan ocean basin and marginal units to the northwest, and were finally thrust over the eastern margin of the Bey Daglari carbonate platform during Miocene time (Poisson, 1977).

An upper limit for the age of remagnetisation can be estimated from the palaeomagnetic data. Kissel and Poisson (1986) reported data from Neogene sediments of the Antalya Basin, which indicate that no rotation has taken place since the Langhian (15 Ma). Four of their sites are of Miocene age. Three of these sites are located in the extreme southeast of the study area, adjacent to the Alanya massif, and may be situated outside of the area of remagnetisation and tectonic rotation. Only one site, of Serravallian–Tortonian age (Middle to Late Miocene), is located within the central Isparta angle. The mean direction of magnetisation at this site ($D = 352°$, $I = 43°$, $\alpha_{95} = 4°$, $n = 8$; Kissel and Poisson, 1986) is significantly different from the regional mean direction of remagnetisation ($D = 326°$, $I = 43°$, $A_{95} = 8.1°$, $N = 10$) determined above. This suggests that remagnetisation and at least 26° of rotation took place before deposition of the sediments sampled at this site.

Studies of Neogene sediments in Western Greece have documented a two-stage clockwise rotation of the western extremity of the Hellenic Arc (Kissel and Laj, 1988). The average rotation rate during the youngest Plio-Quaternary event in this area is 5°/Ma (the earlier Middle Miocene rotation cannot be precisely dated). If we assume that post-Early Miocene rotation in the Isparta angle proceeded at a comparable rate, this would imply that the remagnetisation described here must have been complete at least 5 Ma prior to deposition of the Serravallian–Tortonian sediments sampled by Kissel and Poisson (1986). This suggests that remagnetisation took place before the Tortonian.

It is possible that remagnetisation was associated with Late Palaeocene–Early Eocene emplacement of the Antalya Complex units onto the neighbouring carbonate platforms. This was a relatively local event compared with the scale of thrusting of the large-scale Lycian Nappes from a root zone to the northwest (Poisson, 1977). It is more probable that both remagnetisation and subsequent tectonic rotation were associated with the regional emplacement of the Lycian Nappes during the Miocene. Initial emplacement of these nappes took place in the Burdigalian (Early Miocene). The absence of reversed polarities at the majority of sites reported here suggests that remagnetisation took place during a predominantly normal polarity period. The longest normal polarity chrons in the interval between the Burdigalian and the end of the Serravallian are Chrons 5AC and 5AD (Langhian) and Chron 5C (Burdigalian; Harland et al., 1982). The maximum duration of these chrons is 0.8 Ma.

It is suggested that authigenic magnetite formed in the late Early to Middle Miocene as a result of a short-lived geochemical event related to the migration of orogenic fluids ahead of the advancing Lycian nappes. Rapid remagnetisation events of this kind have been documented previously (Dorobek, 1989; McWhinnie et al., 1990). The newly formed magnetite acquired a remanent magnetisation by grain growth through the critical blocking volume. Subsequently, the region was subjected to a 30° anticlockwise tectonic rotation, driven by the final stages of emplacement of the Lycian nappes in the Late Miocene, regional convergence, and bending of the Aegean arc (as suggested previously by Kissel and Poisson, 1987).

Palaeolatitude of the Isparta angle units

The pervasive Early–Middle Miocene magnetic overprint limits the usefulness of the sampled successions in constraining the palaeolatitude of formation of the Isparta angle units. However, components of magnetisation of assumed primary origin have been recovered from the sequences of the Barla Dag carbonate platform, at sites BA and LE. The inclination data from these sites imply a similar palaeolatitude of approximately 13–17°N for the Barla Dag unit during both the Late Triassic and the Palaeocene/Eocene. This is in agreement with palaeolatitudes calculated from the African polar wander path of Westphal et al. (1986), assuming sites in the Isparta angle were rigidly attached to Africa (Fig. 12). This demonstrates the Gondwanan affinities of this unit. The inclination data from site BA is discussed further below, in relation to the hypothesis of Lauer (1981, 1984).

Additionally, the regional mean inclination of 43° of the remagnetised sites implies a palaeolatitude of 25°N for these units during the Early–Middle Miocene. This is similar to the palaeolatitudes obtained in other Miocene sediments in the Aegean region (e.g., Horner and Freeman, 1983; Kissel et al., 1985; Kissel and Poisson, 1986, 1987). At face value this suggests that the Aegean region as a whole has undergone a northward drift of over 1000 km since the Miocene. This is unreasonable geologically. However, Westphal et al. (1986) have pointed out that low inclination values during the Tertiary are not confined to the

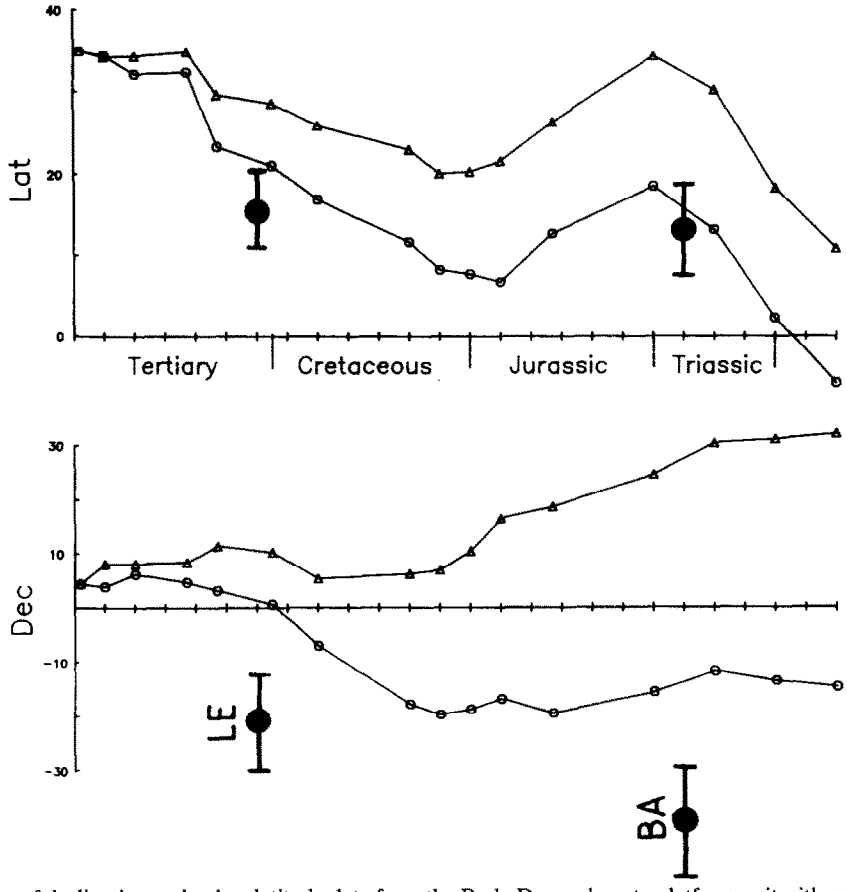


Fig. 12. Comparison of declination and palaeolatitude data from the Barla Dag carbonate platform unit with reference curves given by Westphal et al. (1986) (calculated for a site in the centre of the Isparta angle; latitude = 37°N, longitude = 31°E). △ = Eurasian path. ○ = African path. Note the African affinity of the palaeolatitude data, and the indication of a 30° anticlockwise rotation of the sampled unit with respect to the African reference declination curve.

Aegean area, but have been obtained over a large part of the Mediterranean region, from Iberia to Iran. As yet this phenomenon has no consistent explanation (Westphal et al., 1986; Kissel and Laj, 1988).

Implication for previous palaeomagnetic studies within the Isparta angle

The Early–Middle Miocene remagnetisation documented has several implications for the data previously obtained in the Isparta angle by Kissel and Poisson (1986, 1987) and especially by Lauer (1981, 1984).

The mixed polarity of some of the Bey Daglari carbonate platform sites of Kissel and Poisson (1987) and Lauer (1981) may attest to the primary nature of the magnetisations identified. As discussed above, this does not affect the validity of the remagnetisation hypothesis proposed here.

The main conclusion reached by Kissel and Poisson (1987) was that the Bey Daglari carbonate platform has experienced a ca. 30° anticlockwise rotation since the Middle Miocene (Langhian) at the latest. This rotation is confirmed here. Assuming that at least the mixed polarity sites of Kissel and Poisson (1987) and Lauer (1981) record depositional remanent magnetisations, then the various data suggest that the Bey Daglari massif was not rotated significantly between the Maastrichtian and Early Miocene. It subsequently rotated by 30° anticlockwise in the Late Miocene. This is in agreement with the results from the Barla Dag carbonate platform, which has been shown to have undergone no rotation between the Late Triassic and the Palaeocene/Eocene.

When assessing the geological significance of their data, Kissel and Poisson (1986, 1987) assumed that the rotation of the Bey Daglari carbonate platform died out in the Antalya Complex. However, the data from sites KK and CI of the present study, from the Kemer Zone and Gödene Zone of the southwestern segment of the Antalya Complex respectively, indicate that these units have been rotated along with the Bey Daglari carbonate platform itself.

At Site CI of the present study, within the Late Triassic pillow lavas of the Gödene Zone (Robertson and Woodcock, 1981c) at Çirali (Fig. 2), a negative fold test establishes that the characteristic magnetisation is of post-folding age (Fig. 9). After remagnetisation, this site was subjected to the same anticlockwise tectonic rotation as the Bey Daglari carbonate platform to the west and Kemer Zone off-margin carbonate platform to the east. This has important implications for the data obtained by Lauer (1981) at 11 sites located in the Çalbali Dag lava massif 50 km to the north. After tilt correction of data from these sites, Lauer (1981) obtained mean directions which clustered in the NW quadrant and were of predominantly reversed polarity. However, the pervasive remagnetisation identified in the present study, together with the negative fold test obtained at site CI, suggests that Lauer (1981) was not justified in correcting for structural tilt at these sites. Lauer listed the corrections applied to these data, and so it is possible to recover the in situ site mean directions. Their Fisherian mean is: $D = 336°$, $I = 14°$, K = 16, $\alpha_{95} = 11.6°$, $N = 11$.

This mean direction is comparable to that of the lava blocks sampled at Çirali in the present study (site CI; Table 2). The mean inclination becomes positive, but is not as high as at Çirali. Such variations in inclination might be expected if the two areas experienced an amount of post-remagnetisation differential tilting.

The mean directions identified by Lauer (1981) at these 11 sites are reinterpreted as secondary magnetisations acquired during the Early–Middle Miocene. This interpretation has serious consequences for the hypothesis of Lauer (1981, 1984), in which the Turkish blocks originated close to the equator and subsequently reached their present position by drifting northwards around the Arabian margin (Fig. 1). The proposed equatorial origin for block *B* of Lauer (1981, 1984; Fig. 1) relies heavily on the tilt-corrected inclination data from the Çalbali Dag section. However, there is now good evidence that the Late Triassic lavas of the Gödene Zone do not retain a primary remanence, but only carry secondary components of magnetisation. It is suggested that the inappropriate application of structural tilt corrections to these remagnetised units leads Lauer (1981, 1984)

to position his block *B* so far to the south in the Late Triassic.

Additionally, the Late Triassic carbonates of the Barla Dag massif, sampled at site BA (Fig. 2), probably retain a primary depositional remanence. Although it is clearly dangerous to attach too much importance to data from one site, the mean inclination at site BA can be used to estimate the palaeolatitude of the Barla Dag carbonate platform during the Late Triassic. Assuming an axial geocentric dipole field, the mean tilt corrected inclination of 25° corresponds to a palaeolatitude of 13°N. The apparent polar wander path for Africa given by Westphal et al. (1986) indicates that during the Triassic Africa was moving in a sinistral sense with respect to Eurasia, and with a small northward component of motion. This means that Africa was slightly to the south and to the west of its position in the first of the palaeocontinental reconstructions of Savostin et al. (1986) (for the Lias). With a palaeolatitude of 13°N, the Barla Dag carbonate platform may be placed comfortably within the embayment of the African margin, without resorting to positioning it in the proto-Indian ocean, to the east of the Arabian promontory, as required by Lauer (1981, 1984).

Conclusions

The palaeomagnetic study of both the relatively autochthonous platform units and allochthonous Antalya Complex units of the Isparta angle provides compelling evidence for a widespread Neogene remagnetisation event in the area. This event was possibly triggered by the migration of orogenic fluids ahead of the Lycian nappes during their emplacement on to the carbonate platforms in the Early–Middle Miocene. The secondary magnetisation produced by this event has completely obscured the primary magnetisation at the majority of sites studied.

Subsequent to remagnetisation, a large segment of the Isparta angle area underwent a 30° anticlockwise rotation. This rotation was probably driven by the emplacement of the Lycian Nappes onto the western limb of the Isparta angle and bending of the Aegean arc in the Miocene.

An equatorial origin for one of the three Turkish blocks defined by Lauer (1981, 1984) is not substantiated by this work. Data from the Çalbali Dag section used in the model of Lauer are reinterpreted as resulting from the Early–Middle Miocene overprint. They cannot, therefore, be used to constrain the position of the Gödene Zone in the Late Triassic. Also, new data from the Barla Dag carbonate platform suggest that the units of the present Isparta angle were located close to the northern margin of Gondwana during the Late Triassic.

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